Supplementary Notes S1–S5

S1 The correlated-factor case • S2 Factors, controls and parameters • S3 The completeness matrix • S4 Tier status • S5 The class-level calculation

Companion to "Fuel Anchoring as Reconciliation by Construction: A Greenhouse-Gas Inventory of Road Transport and Mixed Off-Road Mobile Combustion in Pakistan"

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Release provenance. This document describes PakEmis under the period, configuration and source boundaries specified in its methods. The identifiers below link it to the accompanying products and source register.

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Commit	`d8d874622f9a`
Reporting period	July 2015–June 2025 (ten fiscal years) (2015-07-01 to 2025-06-30)
Reference	S_TAN (household survey; TAN nitrogen)
FY2023–24 total	444.61 Mt CO_2e (AR5) under S_TAN, the reference; 449.85 Mt under L_TAN
Release manifest	`ac0ffe93ad9fcd88` (SHA-256 of `outputs/release_manifest.json`, first 16)
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Every table here is generated from the outputs it describes, and a test asserts that the shipped file matches a fresh regeneration. Section and table numbers are those the main paper cites. The machine-readable forms are the CSVs named in each caption.

S1 The Correlated-Factor Case

This table is a retained reference-year diagnostic of two declared fuel-factor dependence settings. It uses the stated legacy seed and parameter implementation; it is distinct from the current common annual conditional ensemble, which shares named physical inputs and pairs years within each current dependence case. The diagnostic does not define a mathematical uncertainty bound.

	Independent	**Correlated**
Sector mean (Mt CO_2e)	41.926	41.922
2.5th percentile	36.869	36.891
97.5th percentile	47.638	47.642
Sector ±%	**12.9**	**12.9**
Road transport ±%	13.2	13.2
Mixed off-road ±%	29.0	28.9

Table S1: Approach 2, 10 000 realisations at seed 42, from `outputs/national_uncertainty_montecarlo.csv`, retained as the legacy reference diagnostic. The central estimate does not move materially—correlation does not bias a sum—and under the common-random-numbers scheme the band does not move either: both treatments print the same width (±12.9 %).

S1.1 What the two scenarios assume

Five sector modules in the national inventory burn fossil fuel, and they draw their carbon-content factors from the same IPCC table entries with the same published uncertainty ranges. The *independent* scenario draws each sector's factor separately. The *correlated* scenario draws one realisation of each fuel's factor and applies it everywhere at once.

The correlated case is physically the more defensible of the two extremes. There is one carbon content of the diesel Pakistan burns, and if the IPCC default is wrong about it, it is wrong in the same direction for road transport, railways, off-road machinery and captive generation simultaneously. Treating the draws as independent asserts that the true value could be at the bottom of its range for railways and the top for road transport at the same moment, which it cannot.

S1.2 Why this sector no longer moves

The retained reference diagnostic rounds to ±13.2 % under either setting—the common-random-numbers rebuild made the correlation scenario a no-op here. Mixed off-road machinery likewise, at ±28.9 % against ±28.9 %.

The difference between the two is instructive. Road transport's uncertainty is dominated by the shared fuel-carbon factor, because everything else about it is comparatively tight: the activity datum is a metered national supply, and CO_2 is 97.5 % of the emissions. When the factor stops being able to average against other sectors' draws, there is very little else left to average, so almost the whole band is exposed. Mixed off-road machinery carries a large independent uncertainty of its own—on the energy share that carves it out, and on the N_2O factor—which continues to average regardless, so the correlation adds proportionally little. Note also why the sector band sits below both category bands in either scenario: the partition uncertainty between road and off-road is negatively coupled and cancels on recombination, while the shared control and carbon-factor terms are positively correlated and survive.

Anchoring is what produces this shape. An inventory that built road transport bottom-up would carry substantial independent uncertainty in its fleet model, and that uncertainty would partly mask the factor correlation. Anchoring removes the fleet uncertainty from the total and leaves the factor exposed, which is a more honest representation of what is actually not known.

S1.3 How to read the sector's band

The earlier sector comparisons below retain their own declared parameter settings and are not the common annual ensemble. The current release samples shared fuel-carbon identities consistently across GHG sources and pairs years within each named dependence case. Perfect persistence and annually varying activity errors are sensitivity assumptions rather than mathematical bounds. Parameter percentiles remain conditional on the road/off-road claimant assignment, transferred factors and admitted sources. Class, sulfur and technology endpoints are structural alternatives and receive no probability.

Beside the correlated-factor case, the sector publishes its own structural-scenario table (`outputs/transport_uncertainty_scenarios_2024.csv`): the boundary cases of the main paper's boundary table (the national-minus-bulk construction, the road-share envelope, the retail non-road cases, the calendar-year construction), the provenance-audit bridges beside it, the CNG heating-value case, the off-road N_2O table choice and share cases, the factor-mix invariance row, the CNG exclusion bound, the fleet-basis energy band, the off-road reporting recode, the class-composition cases at the ends of the declared MJ/km bands and the contrast cases that move light and heavy classes against one another, and two declaration rows—the class split and the air-factor provenance—that are scope decisions rather than samplable parameters. Each carries a conditional parametric interval, an explicit declaration, or a proved invariance, never a structural probability. The deduction and cross-check inputs each carry their own register row: the generator carve-out (survey expenditure over a declared administered price), the off-road energy fraction, and the rail-figure discrepancies—3.8 % year-matched, 11.2 % cross-year—that anchor the controls' declared 10 % (`outputs/transport_uncertainty_inputs_2024.csv`).

S2 Factors, Controls and Parameters with Provenance

Every factor, control total and model parameter the engine reads, rendered from the same CSV files the code reads (`data/emission_factors.csv`, `data/transport_params.csv`, `data/activity/transport/emep_tier1_road_air.csv`, `data/activity/transport/emep_class_map.csv`, `data/activity/transport/fuel_sulfur_by_year.csv` and the superseded `data/activity/transport/emission_factors_gkm.csv`) so the table cannot drift from the calculation.

Table S2: Emission factors, fuel controls and parameters. The `fuel_supply` rows are the OCAC Road customer-class controls; the `supply_legacy` rows are the superseded PES-derived pair, retained for the `legacy_pes_control` scenario with their provenance error documented, and `supply_pes12` is that note's arithmetic done correctly at its (wrong) FY2025 vintage. The household-generator carve-out is relayed from the residential sector and the full claimant structure appears in `outputs/transport_fuel_ledger_2024.csv`.

Group	Item	Value	Unit	±%	Source
fuel_supply	Gas/Diesel Oil	5551.44	kt/yr	10	OCAC Sectoral Sales of Energy Products FY2023-24, Transport/Road row: HSD 5,551,442 t (data/activity/transport/ocac_sectoral_annual.csv)
fuel_supply	Motor Gasoline	7209.41	kt/yr	10	OCAC Sectoral Sales of Energy Products FY2023-24, Transport/Road row: MS 7,104,484 + HOBC 104,926 t (data/activity/transport/ocac_sectoral_annual.csv)
fuel_supply_legacy	Gas/Diesel Oil	8820	kt/yr	15	Pakistan Economic Survey 2024-25 energy chapter Fig 14.4 (source OCAC): HSD 4,961.0 kt for July-March FY2025, annualised by 12/9
fuel_supply_legacy	Motor Gasoline	10162	kt/yr	15	Pakistan Economic Survey 2024-25 energy chapter Fig 14.4 (source OCAC): MS 5,512.5 plus HOBC 203.5 kt for July-March FY2025, annualised by 12/9
fuel_supply_pes12	Gas/Diesel Oil	6614.67	kt/yr	15	Pakistan Economic Survey 2024-25 energy chapter Fig 14.4 (source OCAC): HSD 4,961.0 kt for July-March FY2025, annualised by 12/9
fuel_supply_pes12	Motor Gasoline	7621.33	kt/yr	15	Pakistan Economic Survey 2024-25 energy chapter Fig 14.4 (source OCAC): MS 5,512.5 plus HOBC 203.5 kt for July-March FY2025, annualised by 12/9
ncv	Gas/Diesel Oil	43	TJ/Gg	3	IPCC 2006 GL Vol 2 Ch 1 Table 1.2
ncv	Motor Gasoline	44.3	TJ/Gg	4	IPCC 2006 GL Vol 2 Ch 1 Table 1.2
fuel_co2	Gas/Diesel Oil	74100	kg CO2/TJ	2	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.1 (Road Transport)
fuel_co2	Motor Gasoline	69300	kg CO2/TJ	2	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.1 (Road Transport)
fuel_co2	Natural Gas	56100	kg CO2/TJ	4	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.1 (Road Transport)
fuel_ch4	Gas/Diesel Oil	3.9	kg CH4/TJ	60	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (diesel)
fuel_ch4	Motor Gasoline	33	kg CH4/TJ	60	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (uncontrolled petrol light-duty)
fuel_ch4	Natural Gas	92	kg CH4/TJ	80	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (natural gas vehicles)
fuel_n2o	Gas/Diesel Oil	3.9	kg N2O/TJ	70	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (diesel)
fuel_n2o	Motor Gasoline	3.2	kg N2O/TJ	70	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (uncontrolled petrol light-duty)
fuel_n2o	Natural Gas	3	kg N2O/TJ	70	IPCC 2006 GL Vol 2 Ch 3 Table 3.2.2 (natural gas vehicles)
off-road_co2	Gas/Diesel Oil	74100	kg CO2/TJ	2	IPCC 2006 GL Vol 2 Ch 3 Table 3.3.1 (Off-road mobile sources and machinery; Agriculture /Diesel)
off-road_ch4	Gas/Diesel Oil	4.15	kg CH4/TJ	130	IPCC 2006 GL Vol 2 Ch 3 Table 3.3.1 (Off-road; Agriculture /Diesel)
off-road_n2o	Gas/Diesel Oil	28.6	kg N2O/TJ	150	IPCC 2006 GL Vol 2 Ch 3 Table 3.3.1 (Off-road; Agriculture /Diesel)
param	activity_uncert	25	percent	0	IPCC 2006 GL Vol 2 Ch 3 s3.2.1
param	cng_activity_uncert	10	percent	0	OGRA metered supply to licensed CNG stations, reported in two independent tables of the same edition that agree

Table S2 continued

Group	Item	Value	Unit	±%	Source
param	cng_hhv_btu_per_scf	1000	Btu per standard cubic foot (gross)	8	SNGPL and SSGC bill in MMBTU at a nominal 1,000 Btu/scf; OGRA reports CNG supply in MMCFD
param	cng_gross_to_net	0.9	fraction	3	IPCC 2006 GL Vol 2 Ch 1 s1.4.1.2: net calorific value is about 10 percent below gross for natural gas
param	suv_share_start	0.05	fraction	50	Reconstructed from the Punjab workbook's Classified sheet (5.0% in 2003-2005)
param	suv_share_end	0.2	fraction	50	Reconstructed from the Punjab workbook's Classified sheet (20.0% in 2023)

Class	Weibull β	Median life (yr)
2W	3.1	14
3W	2	15
4W1	3	16
4W2	2.5	15
4WT	3	12
LDV	3	20
BUS	3	12
HDV	3.8	18
NRV	4	22

Table S3: Survival-model parameters by class (Naveed 2024, §3.7.3 Table 5, adopting the Guttikunda 2024 high-survival scenario); β at ±25%, life at ±20%.

Process	Category	Fuel	Value	min	max	Unit
exhaust CO	PC	Petrol	48.36	8.6	338.69	g per kg fuel
exhaust NMVOC	PC	Petrol	7.75	1.15	176.89	g per kg fuel
exhaust NO_x	PC	Petrol	3.98	0.44	30.49	g per kg fuel
exhaust $PM_{2.5}$	PC	Petrol	0.02	0.004	0.042	g per kg fuel
exhaust NH_3	PC	Petrol	0.84	0.02	1.93	g per kg fuel
exhaust CO	PC	Diesel	2.41	0.52	11.23	g per kg fuel
exhaust NMVOC	PC	Diesel	0.51	0.02	3.26	g per kg fuel
exhaust NO_x	PC	Diesel	11.77	7.79	14.84	g per kg fuel
exhaust $PM_{2.5}$	PC	Diesel	0.78	0.002	4.772	g per kg fuel
exhaust NH_3	PC	Diesel	0.02	0.01	0.04	g per kg fuel
exhaust CO	PC	LPG	58.22	14.56	139.93	g per kg fuel
exhaust NMVOC	PC	LPG	9.43	1.48	20.85	g per kg fuel
exhaust NO_x	PC	LPG	5.48	0.9	41.45	g per kg fuel
exhaust $PM_{2.5}$	PC	LPG	0.03	0.012	0.042	g per kg fuel
exhaust NH_3	PC	LPG	1.02	0.03	1.85	g per kg fuel
exhaust CO	LCV	Petrol	118.7	11.75	356.95	g per kg fuel
exhaust NMVOC	LCV	Petrol	7.42	0.67	53.99	g per kg fuel
exhaust NO_x	LCV	Petrol	5.93	0.33	37.31	g per kg fuel
exhaust $PM_{2.5}$	LCV	Petrol	0.02	0.005	0.034	g per kg fuel
exhaust NH_3	LCV	Petrol	0.77	0.02	2.04	g per kg fuel
exhaust CO	LCV	Diesel	6.81	4.82	21.77	g per kg fuel
exhaust NMVOC	LCV	Diesel	1.23	0.48	2.86	g per kg fuel
exhaust NO_x	LCV	Diesel	13.48	10.43	27.86	g per kg fuel
exhaust $PM_{2.5}$	LCV	Diesel	1.22	0.002	4.772	g per kg fuel
exhaust NH_3	LCV	Diesel	0.01	0.01	0.03	g per kg fuel
exhaust CO	HDV	Diesel	6.1	2.79	13.8	g per kg fuel
exhaust NMVOC	HDV	Diesel	0.9	0.13	7.01	g per kg fuel
exhaust NO_x	HDV	Diesel	25.95	10.48	47.22	g per kg fuel
exhaust $PM_{2.5}$	HDV	Diesel	0.55	0.015	2.448	g per kg fuel
exhaust NH_3	HDV	Diesel	0.02	0.01	0.11	g per kg fuel
exhaust CO	HDV	CNG	3.98	2.2	15.12	g per kg fuel
exhaust NMVOC	HDV	CNG	0.14	0.06	0.39	g per kg fuel
exhaust NO_x	HDV	CNG	17.1	9.67	29.73	g per kg fuel
exhaust $PM_{2.5}$	HDV	CNG	0.02	0.019	0.036	g per kg fuel
exhaust CO	L	Petrol	244.88	8.46	588.11	g per kg fuel
exhaust NMVOC	L	Petrol	92.74	6.47	323.68	g per kg fuel
exhaust NO_x	L	Petrol	7.23	0.89	22.06	g per kg fuel
exhaust $PM_{2.5}$	L	Petrol	0.51	0.058	6.599	g per kg fuel
exhaust NH_3	L	Petrol	0.05	0.04	0.11	g per kg fuel
bc fraction BC	PC	Petrol	0.12	–	–	fraction of PM2.5
bc fraction BC	LCV	Petrol	0.05	–	–	fraction of PM2.5
bc fraction BC	PC	Diesel	0.57	–	–	fraction of PM2.5
bc fraction BC	LCV	Diesel	0.55	–	–	fraction of PM2.5
bc fraction BC	HDV	Diesel	0.53	–	–	fraction of PM2.5
bc fraction BC	L	Petrol	0.11	–	–	fraction of PM2.5
evaporative NMVOC	PC	Petrol	14.6	10.5	19.2	g per vehicle day
evaporative NMVOC	LCV	Petrol	22.2	9.9	27.9	g per vehicle day
evaporative NMVOC	L	Petrol	7.5	5.8	9.1	g per vehicle day
tyre brake wear TSP	L	–	0.0083	0.0064	0.0103	g per vkm
tyre brake wear PM_{10}	L	–	0.0064	0.0047	0.0081	g per vkm
tyre brake wear $PM_{2.5}$	L	–	0.0034	0.0026	0.0042	g per vkm
tyre brake wear TSP	PC	–	0.0229	0.0083	0.0369	g per vkm
tyre brake wear PM_{10}	PC	–	0.0184	0.0067	0.0297	g per vkm
tyre brake wear $PM_{2.5}$	PC	–	0.0093	0.0034	0.015	g per vkm
tyre brake wear TSP	LCV	–	0.03427	0.019	0.045	g per vkm
tyre brake wear PM_{10}	LCV	–	0.0271	0.0148	0.0351	g per vkm
tyre brake wear $PM_{2.5}$	LCV	–	0.0139	0.0076	0.018	g per vkm
tyre brake wear TSP	HDV	–	0.0777	0.0462	0.1318	g per vkm
tyre brake wear PM_{10}	HDV	–	0.059	0.05	0.095	g per vkm
tyre brake wear $PM_{2.5}$	HDV	–	0.0316	0.0281	0.0541	g per vkm
road wear TSP	L	–	0.006	0.0036	0.0081	g per vkm
road wear PM_{10}	L	–	0.003	0.0018	0.0041	g per vkm
road wear $PM_{2.5}$	L	–	0.0016	0.001	0.0022	g per vkm
road wear TSP	PC	–	0.015	0.009	0.0203	g per vkm
road wear PM_{10}	PC	–	0.0075	0.0045	0.0101	g per vkm
road wear $PM_{2.5}$	PC	–	0.0041	0.0024	0.0055	g per vkm
road wear TSP	LCV	–	0.021	0.009	0.0253	g per vkm
road wear PM_{10}	LCV	–	0.0105	0.0045	0.0123	g per vkm
road wear $PM_{2.5}$	LCV	–	0.0057	0.0024	0.007	g per vkm
road wear TSP	HDV	–	0.076	0.0456	0.1026	g per vkm
road wear PM_{10}	HDV	–	0.038	0.0228	0.0513	g per vkm
road wear $PM_{2.5}$	HDV	–	0.0205	0.0123	0.0277	g per vkm

Table S4: EMEP/EEA guidebook 2023 Tier 1 road-transport factors, transcribed from the held chapter PDFs (road transport Update 2025 Tables 3-5/3-6/3-7/3-11; gasoline evaporation Table 3-1 at the 20–35 °C band; tyre-and-brake and road-surface wear Tables 3-1/3-2), each cell re-checked against the source text by `scripts/check_transcriptions.py`. min/max span uncontrolled to 2010-current technology and are carried as each row's uncertainty. Exhaust PM is $PM_{2.5}$ by the

Class	Fuel	Tier 1 category	Tier 1 fuel	Declared proxy
2W	Petrol	L	Petrol	–
3W	Petrol	L	Petrol	–
4W1	Petrol	PC	Petrol	–
4W2	Petrol	PC	Petrol	–
4WT	Petrol	PC	Petrol	–
BUS	Petrol	LCV	Petrol	declared proxy: no Tier 1 petrol bus/HDV factor exists; near...
LDV	Petrol	LCV	Petrol	–
4W2	Diesel	PC	Diesel	–
BUS	Diesel	HDV	Diesel	–
HDV	Diesel	HDV	Diesel	–
LDV	Diesel	LCV	Diesel	–
2W	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
3W	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
4W1	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
4W2	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
4WT	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
LDV	CNG	PC	LPG	declared proxy: spark-ignition gaseous fuel; the guidebook h...
BUS	CNG	HDV	CNG	the guidebook's own CNG (Buses) row
HDV	CNG	HDV	CNG	the guidebook's own CNG (Buses) row - the nearest heavy-duty...

Table S5: The declared bridge from the fleet model's classes to the Tier 1 categories (`emep_class_map.csv`): a cell without a published Tier 1 factor in the held chapters takes a named proxy, never a silent one.

Year	Fuel	ppm	lo	hi	Basis
2016	Diesel	5000	500	10000	pre-Euro-II circulation at 0.5 percent sulfur
2017	Diesel	5000	500	10000	transition year - the January 2017 Euro-II import mandate falls ...
2018	Diesel	500	500	5000	Euro-II 500 ppm import specification
2019	Diesel	500	500	5000	Euro-II 500 ppm import specification
2020	Diesel	500	500	5000	Euro-II 500 ppm import specification
2021	Diesel	500	10	5000	Euro-V import mandate January 2021 falls mid-fiscal; domestic ou...
2022	Diesel	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2023	Diesel	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2024	Diesel	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2016	Petrol	500	150	1000	Euro-2-era petrol specification ceiling
2017	Petrol	500	150	1000	Euro-2-era petrol specification ceiling
2018	Petrol	500	150	1000	Euro-2-era petrol specification ceiling
2019	Petrol	500	150	1000	Euro-2-era petrol specification ceiling
2020	Petrol	500	150	1000	Euro-2-era petrol specification ceiling
2021	Petrol	500	10	1000	Euro-V petrol import mandate August 2020; domestic output unchan...
2022	Petrol	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2023	Petrol	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2024	Petrol	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2025	Diesel	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...
2025	Petrol	500	10	1000	supply mixes Euro-V imports with domestic refinery output; conse...

Table S6: The declared road-fuel sulfur regime by fiscal year (`fuel_sulfur_by_year.csv`), behind the stoichiometric SO_2; declared from the cited import-mandate record, not measured at the pump (gap `road-fuel-sulfur`). CNG carries no sulfur row: pipeline gas is desulfurised and the absence is an absence, not a zero.

Class	**Fuel**	NO_x	CO	NMVOC	$PM_{2.5}$	PM_{10}	SO_2	NH_3
2W	Petrol	0.1	12	2.5	0.05	0.06	0.024	0.01
2W	CNG	0.08	8	1.2	0.01	0.012	0	0.01
3W	Petrol	0.15	15	4	0.06	0.072	0.032	0.01
3W	CNG	0.12	10	1.5	0.01	0.012	0	0.01
4W1	Petrol	0.5	25	3.5	0.03	0.036	0.055	0.1
4W1	CNG	0.4	18	2	0.01	0.012	0	0.05
4W2	Petrol	0.45	20	2.5	0.04	0.048	0.05	0.08
4W2	Diesel	0.8	2.5	0.3	0.2	0.24	0.4	0.02
4W2	CNG	0.4	15	1.8	0.01	0.012	0	0.02
4WT	Petrol	0.5	25	3.5	0.03	0.036	0.055	0.1
4WT	CNG	0.4	18	2	0.01	0.012	0	0.05
LDV	Petrol	0.45	20	2.5	0.04	0.048	0.05	0.08
LDV	Diesel	1	3	0.4	0.25	0.3	0.5	0.02
LDV	CNG	0.45	12	1.5	0.01	0.012	0	0.02
HDV	Diesel	6	10	1	1	1.2	2	0.05
HDV	CNG	3.5	8	0.8	0.1	0.12	0	0.05
BUS	Petrol	0.6	12	2	0.05	0.06	0.05	0.05
BUS	Diesel	7	8	0.8	0.8	1	2.5	0.05
BUS	CNG	4	6	0.6	0.08	0.1	0	0.05
NRV	Diesel	5	8	1	0.8	1	2	0.02

Table S7: SUPERSEDED per-kilometre factors (g/km), retained for the old-vs-new comparison in the air appendix: these are transcriptions of the Punjab emissions workbook's values, for which no primary citation was ever located, and the EMEP/EEA Tier 1 tables above replaced them as the layer's factor basis. Nothing canonical reads this table any longer. NMVOC is non-methane VOC; NO_x is NO_2-equivalent mass, the inventory-wide basis. Rows the fleet model never produces (all-zero) are omitted.

S3 The Transport-Boundary Completeness Matrix

Every category–gas combination on the transport boundary, with its IPCC notation key and the justification the engine carries. The boundary spans four reporting codes deliberately: the diesel this sector's residual class and unresolved claimants actually belong to sits in 1.A.2, and the vehicle fleet's refrigerant leakage in IPPU 2.F.1. Generated from the same completeness table the national inventory publishes, which is itself synced to the gap register by test.

Table S8: The transport-boundary completeness matrix, generated from the engine (`outputs/completeness.csv`, explicit allowlist). E = estimated, IE = included elsewhere, NE = not estimated (each NE row has a gap-register entry stating what would resolve it), NO = not occurring.

Code	Category	Key	Gases	Justification (abridged)
1.A.3.b	Road transport	E	CO2/CH4/ N2O	ANCHORED ON FUEL, not on the fleet model: the OCAC FY2023-24 Road customer-class rows (petrol MS+HOBC and HSD - bulk industry, power, government and railway diesel excluded by the source's own partition), net of the household-generator petrol carve-out relayed from the residential sector, plus CNG from OGRA's metered sector gas on its own activity basis. The railway figure is an independent cross-check (Year Book FY2023-24 row vs OCAC Railways row, 3.8 percent apart), not a deduction. outputs/transport_ fuel_ ledger_ 2024.csv is the executable two-tier claimant structure. KNOWN FLAW: the class split rides vehicle-kilometres, over-weighting motorcycles; the total is unaffected because it is set by the fuel rows. Retail-bought agricultural-stationary, captive and other non-road diesel remains inside the Road row (the ledger's UNRESOLVED rows), biasing 1.A.3.b high conditional on the other...
1.A.3.b	Electric road vehicles	NO	CO2/CH4/ N2O	Zero direct combustion by definition, and the registered electric fleet is negligible in the 2022 registration data. Electricity-related emissions are counted at the power station in 1.A.1.a like all consumption - the Orange Line convention. Recorded so the road scope statement has a boundary row rather than an implicit omission.
1.A.3.c	Railways	E	CO2/CH4/ N2O	LOCOMOTIVE TRACTION ONLY under the re-based reference (0.38939 Mt): the Year Book detail table splits locomotive from other-than-locomotive fuel, and the measured non-traction share books at its own category row below. Pakistan Railways Year Book metered fuel consumption x IPCC Vol 2 Ch 3 Table 3.4.1, the RAIL table - its N2O factor is 7.3 times the road value. Railway diesel is its own OCAC customer-class row, outside the road control by the source's structure, and the Year Book figure serves as an independent cross-check on that row (3.8 percent apart at the matched year) rather than a deduction. The mainline network has been diesel-only since about 2010 (electric energy NIL from the 2010-2015 quinquennium in the Year Book's own historical table), so there is no scope-2 stream in the inventory years; urban electric rail (Lahore Orange Line) is outside the operator, has zero direct 1A3...
1.A.3.a	Domestic aviation	E	CO2/CH4/ N2O	LTO cycles from CAA movement statistics x ICAO fuel-burn tables, full-flight for domestic. International bunkers are a memo item ESTIMATED at full flight by the source-row-conserving JP-1 bridge (550,670 t of commercial JP-1, 1.740 Mt CO2, outputs/aviation_ bridge_ 2024.csv); the 210.4 kt international-LTO figure is a separate local-air product, not the bunkers estimate.

Table S8 continued

Code	Category	Key	Gases	Justification (abridged)
1.A.3.d	Water-borne navigation - domestic	NE	CO2/CH4/ N2O	SCREENED, not estimated: the two blockers the earlier row named - a marine factor choice and a domestic/foreign boundary review - are made in sectors/transport/navigation_screen.py, and every held OCAC edition's Domestic Bunkers row is converted in outputs/transport_navigation_screen_ 2024.csv (3,054 t HSD in the FY2023-24 reference year, 0.0098 Mt CO2e on IPCC Tables 3.5.2/3.5.3). It stays NE for two stated reasons: the reference-year totals are frozen through the release-candidate series, and the ten held editions span 32 to 8,342 t with no trend - an activity that volatile needs the port-side series before it can carry an E. See marine-domestic.
1.A.4.c	Off-road machinery (tractors and plant)	E	CO2/CH4/ N2O	Partitioned from the OCAC Road-row diesel control by the off-road share of vehicle-kilometres - the same normalisation that splits the on-road classes, so the two cannot claim the same tonne. IPCC Vol 2 Ch 3 Table 3.3.1 factors, whose N2O is 7.3 times the road value (and identical between the table's Agriculture and Industry rows, so the class's mixed composition is factor-invariant under the correct table). KNOWN FLAW: the class is the source workbook's residual bucket as well as its tractor class, is modelled as 100% diesel, and its exact share is priced by the offroad_share scenarios.
1.A.4.c	Stationary agriculture/forestry and railway workshops	IE	CO2/CH4/ N2O	INCLUDED ELSEWHERE, not unestimated - the honest key this row carried wrong as NE: the fuel IS estimated, just in the wrong category at the wrong factors. Stationary farm plant (tube wells, threshing engines) burns diesel that remains inside the 1.A.3.b road control total and is reported there at road factors - an UNRESOLVED claimant row of outputs/transport_fuel_ledger_2024.csv; OCAC publishes an agriculture row that would split it out (transport-hsd-end-use). Railway workshop, power-house, pump and track-machine fuel is MEASURED at 1,599 thousand litres (1.21 percent of the p.163 detail total) by pkinv.sectors.railways.reconcile and, since the rail re-base, reported at its own category row (1.A.2/1.A.4.c unresolved) rather than inside 1.A.3.c; the source-level split below that figure is still not published. See stationary-farm-rail-workshop and rail-yearbook-detail.
1.A.2	Captive-generation diesel and furnace oil	IE	CO2/CH4/ N2O	Industrial self-generation liquids are IPCC 1.A.2 and are INCLUDED ELSEWHERE in two places. Bulk purchases (the channel a captive plant of any size buys through) sit inside the ESTIMATED 1.A.2 industrial liquids - OCAC's Total Industry customer-class rows (src/pkinv/sectors/industry/data.py), which the transport fuel ledger excludes from the road control by construction (its RESOLVED-OBSERVED Industry claimant row). Only retail-pump captive diesel remains inside the 1.A.3.b Road control at road factors - the UNRESOLVED claimant row 'Captive generation (retail)' of outputs/transport_ fuel_ledger_2024.csv, declared as biasing road transport high. Captive GAS self-generation is a separate, estimated category (computed by the power module, reported under industry through data/stream_reporting_sector.csv); this row is its liquid-fuel sibling. See transport-hsd-end-use.

Table S8 continued

Code	Category	Key	Gases	Justification (abridged)
1.A.2	Construction and industrial mobile machinery (1A2gvii)	IE	CO2/CH4/ N2O	IPCC books construction and industrial off-road machinery under 1.A.2 (gvii), NOT under 1.A.4.c - a mapping this inventory previously overgeneralised. The fuel is reported inside the Mixed off-road (1A4c/1A2) category, whose class is the source workbook's residual bucket as well as its tractor class, so whatever construction plant the residual contains is estimated there at Table 3.3.1 factors and allocated on a cropland proxy that is wrong for it. Splitting the residual needs equipment stock data no source here provides.
2.F.1	Mobile air-conditioning (road vehicles)	NE	HFC	Refrigerant leakage from vehicle air-conditioning is IPPU 2.F.1, not combustion, so it is outside the fuel anchor by construction - but it is a real transport-fleet emission this inventory does not yet report anywhere: the 2.F species-aware bank engine is implemented and the Ozone Secretariat's 2024 consumption control is loaded, but the mobile-air-conditioning application split has not been reconciled here. See transport-mobile-ac-hfc.
(air)	Air pollutants - road transport	E	PM2.5/ PM10/CO/ NOx/ NMVOC/ SO2/NH3	EMEP/EEA 2023 Tier 1 factors per class and process on the OCAC road fuel the greenhouse-gas side burns, so modelled fuel equals sold fuel: exhaust (g/kg fuel), petrol evaporation (g/vehicle/day), tyre and brake wear and road-surface wear (g/vkm). SO2 from declared fuel sulfur by year, a policy-date assumption rather than an assay. Resuspended road dust is not estimated. Off-road machinery is 1A4c, outside this layer. The factors describe an EU-27 2010 fleet, not Pakistan's.
(air)	Air pollutants - road transport cold-start excess	IE	CO/ NMVOC/ NOx/PM2.5	Included in the exhaust row: the Tier 1 factors are derived from the guidebook's hot plus cold Tier 3 method for a typical EU-27 2010 fleet (2023 guidebook 1.A.3.b, section 3.2.2), so warm-up excess is inside them at European trip and temperature conditions. No Pakistan-specific cold-start treatment is applied; that would need trip counts, trip lengths and ambient temperature. See transport-cold-start.
(air)	Air pollutants - road transport evaporative NMVOC	E	NMVOC	Estimated by the air program's Tier 1 refactor: the 1.A.3.b.v Gasoline evaporation chapter was acquired (sources/EMEP/) and its per-vehicle-day factors (20-35 C band) ride the fleet model's own active petrol counts as separate evaporative atoms in pkinv.sectors.transport.air_pollutants. The year-round hottest-band assumption is the layer's declared limitation. Gap transport-evaporative-nmvoc is CLOSED.
(air)	Air pollutants - road transport non-exhaust PM (tyre, brake and road...	E	TSP/PM10/ PM2.5	Estimated by the air program's Tier 1 refactor: the 1.A.3.b.vi-vii chapters were acquired (sources/EMEP/) and their g/vkm factors ride the same survival-weighted vehicle-kilometres the fuel split rides, as separate tyre_brake_wear and road_ wear atoms - the coarser TSP/PM10 strata beside, never summed into, the PM2.5 core. Road-dust resuspension stays out. Gap transport-nonexhaust-pm is CLOSED.

S4 Tier Status by Component

Tier status differs by component, and two of the sector's defining pieces – the class split and the spatial allocation – are country-specific allocations that IPCC's tier ladder does not classify at all. One word for the whole sector would either overstate the factors or understate the activity.

Component	Status	Basis
Road CO_2 (petrol, diesel)	Tier 1, fuel-based	OCAC Road customer-class rows × IPCC Table 3.2.1 defaults (69,300 / 74,100 kg/TJ). The activity is country-specific and the anchor is exact by test; the factors are defaults, so Tier 1 despite the country-specific control.
Road CH_4, N_2O	Tier 1	Table 3.2.2 uncontrolled defaults on the same fuel energy. Tier 2 needs the technology distribution (emission-control classes) this fleet model does not resolve.
CNG stream	Tier 1, metered activity	OGRA metered sector gas × Table 3.2.2 CNG defaults; the heating value is the declared assumption (900–1,000 Btu/scf), priced as a structural scenario.
Class split	Country-specific allocation (not a tier)	Registration-derived survival-weighted vehicle-km shares. IPCC tiers classify emission estimation, not allocation; calling the sector Tier 2 on the strength of this split would overstate it. No independent validation exists for the split (see the main paper's limitations).
Mixed off-road (1A4c/1A2)	Tier 1	Diesel share by vehicle-km partition × Table 3.3.1 defaults (N_2O 28.6 kg/TJ vs the road table's 3.9; the choice is pinned by test). The class absorbs the source workbook's residual bucket, so its composition is soft - though factor-invariant under Table 3.3.1, whose Agriculture and Industry diesel rows are identical.
Air-pollutant layer	EMEP/EEA 2023 Tier 1	Per class and process on the OCAC road fuel: exhaust in g/kg fuel (hot and cold running inside the factor, for an EU-27 2010 fleet), petrol evaporation, tyre and brake wear and road-surface wear. SO_2 from declared fuel sulfur by year. Resuspended road dust is not estimated; no Pakistan-specific cold-start or fleet-technology treatment is applied.
Spatial allocation	Country-specific proxies (not a tier)	Petroleum streams ride provincial fuel sales from OCAC's monthly area editions (petrol and diesel separately); CNG rides its own fleet vehicle-kilometre proxy and the mixed off-road class rides cropland. Reconciliation to the national control is exact by construction; the superseded registration proxy is retained as a published failure diagnostic.

Table S9: Tier status by component. Generated; the factor values quoted are asserted against `data/emission_factors.csv` at generation time.

S5 The Class-Level Calculation, in Full

Every quantity that determines the class split, including the declared energy intensities that weight it. The within-fuel weights are the $s_{c,f}$ of the main paper's methods section: vehicle-kilometres times MJ/km, normalised within each fuel over *all* classes including off-road, which is the deliberate normalisation that lets the two categories partition one diesel quantity. Because the weights sum to one by construction, the intensities move composition and never magnitude. Machine-readable: `outputs/transport_class_parameters_2024.csv`.

Class		Active fleet	Registered	A/R	km/veh	β	Life	Fuel split	Cells	Capped
2W	Two-wheelers	15,616,588	20,313,397	0.7688	12,300	3.1	14	P 1	95	11
3W	Three-wheelers	557,016	710,090	0.7844	57,000	2	15	C 0.55; P 0.45	79	14
4W1	Cars (small)	1,935,561	2,287,560	0.8461	11,300	3	16	C 0.12; P 0.88	95	39
4W2	Cars (large/SUV)	12,020	12,257	0.9807	11,300	2.5	15	D 0.65; P 0.35	23	7
4WT	Taxis	56,404	70,878	0.7958	57,000	3	12	C 0.52; P 0.48	78	41
LDV	Light duty	402,304	420,121	0.9576	32,400	3	20	D 0.3; P 0.7	94	48
BUS	Buses	132,571	166,282	0.7973	45,600	3	12	C 0.28; D 0.68; P 0.04	93	41
HDV	Heavy duty	240,675	246,354	0.9769	23,300	3.8	18	C 0.02; D 0.98	91	47
NRV	Off-road (mixed)	941,248	959,174	0.9813	1,900	4	22	D 1	93	36

Table S10: Class parameters, FY2023–24 (registration panel 2023, a declared borrowed shape): survival-weighted active fleet against registered stock, annual kilometres, Weibull parameters (Naveed 2024, adopting Guttikunda 2024), fuel splits, and the district cells the Weibull cap binds in.

Class	Fuel	Vehicle-km/yr	MJ/km	Energy (MJ)	Weight
2W	Petrol	1.921e+11	0.85	1.633e+11	0.609992
3W	CNG	1.746e+10	1.7	2.969e+10	0.453040
3W	Petrol	1.429e+10	1.4	2.000e+10	0.074730
4W1	CNG	2.625e+09	2.5	6.562e+09	0.100136
4W1	Petrol	1.925e+10	2.35	4.523e+10	0.168986
4W2	Diesel	8.829e+07	3.8	3.355e+08	0.002378
4W2	Petrol	4.754e+07	3.6	1.711e+08	0.000639
4WT	CNG	1.672e+09	2.5	4.180e+09	0.063784
4WT	Petrol	1.543e+09	2.35	3.627e+09	0.013549
BUS	CNG	1.693e+09	14	2.370e+10	0.361645
BUS	Diesel	4.111e+09	12.5	5.138e+10	0.364136
BUS	Petrol	2.418e+08	8.5	2.055e+09	0.007679
HDV	CNG	1.122e+08	12.5	1.402e+09	0.021395
HDV	Diesel	5.496e+09	11	6.045e+10	0.428387
LDV	Diesel	3.910e+09	4.2	1.642e+10	0.116386
LDV	Petrol	9.124e+09	3.65	3.330e+10	0.124424
NRV	Diesel	1.788e+09	7	1.252e+10	0.088713

Table S11: The within-fuel weights $s_{c,f}$, with the kilometres and the declared MJ/km midpoints they are built from. The two-wheeler petrol weight is 0.6100: 61.0% of the petrol control. On distance alone it would be 0.8119, and the difference between those two numbers is the whole correction – a motorcycle-kilometre and a car-kilometre are not the same quantity of petrol. The intensities are declared engineering judgement, not Pakistani measurement, which is why they are confined to a normalisation that cannot move the sector total.

The off-road partition share, with its actual inputs – energy, not distance, because the same normalisation that splits the on-road classes sets this share:

$$s_{\text{offroad}} = \frac{1.252e+10}{1.252e+10 + 1.286e+11} = 0.08871,$$

in megajoules of declared fleet energy, which carves 21 177 TJ from the Road-row diesel control. The numerator is the off-road class's 941,248 active vehicles at 1900 km each at 7 MJ km^{-1}; on distance alone the share was 0.09998, and it falls because off-road machinery burns LESS per kilometre than the on-road diesel fleet average (7 MJ/km against roughly 9.5, a mean dominated by buses and heavy goods vehicles) – weighting by energy shrinks the off-road share and moves diesel back to the road category. The `offroad_share_5pct` and `15pct` scenarios price the share's softness.